

JOYITA FARUK

Website: joyitaf.github.io

Mail: joyitafaruk@gmail.com

linkedin.com/in/joyita-faruk

github.com/joyitaf

EDUCATION

Bachelor of Science | *Computer Science* | *CGPA 3.52/4.0*
Independent University, Bangladesh

Sep. 2019 – Aug 2024
Dhaka, Bangladesh

WORK EXPERIENCE

Junior Data Scientist
SimpliSolve

Jun 2025 – Current
Texas, US (Remote)

- Understanding and reproducing state-of-the-art time series prediction algorithms
- Conducting thorough exploratory data analysis, discerning patterns and insights
- Writing efficient and robust GPU-compatible Python code to implement algorithms
- Deploying developed codes using cloud services such as AWS

Genomics Intern
Genereveal

Feb 2025 – Current
Montana, US (Remote)

- Working on human DNA sequence data, exploring disease associations
- Leveraging Slurm-based HPC for large-scale genomic data processing
- Analyzing genomic data via alignment, variant calling, and tools like BWA, GATK, and SAMtools
- Developing and optimizing bash scripts to automate workflows

Editor-in-Chief
Kitchen Sink, A Literary Journal

Jun 2021 – Jun 2022
Dhaka, Bangladesh (Remote)

- Co-founded and led content management for an online poetry journal during the COVID-19 lockdown
- Built and nurtured a community of over 2,500 followers through curated content
- Organized and hosted a 3-hour live poetry event with 17 performers and an audience of over 100 attendees

RESEARCH

Kidney Abnormality Detection Using Segmentation-Guided Classification on CT Images

Published in: International Conference on Machine Learning and Cybernetics (ICMLC), Miyazaki, Japan.
Pages 414–419. DOI: 10.1109/ICMLC63072.2024.10935113

Kidney Abnormal Region Localization on 3D CT Images Using ROI-Guided Segmentation

Journal paper in progress

Quantum-Hybridized Cloud Integration for Next-Generation Security and Performance

Journal paper in progress

Cloud-Based Vaccine Tracking System: A Study on Confidentiality and Performance Trade-offs

Journal paper in progress

SKILLS

Programming Languages: Python, Java, C++
Machine Learning & AI: Keras, TensorFlow, PyTorch
Cloud & DevOps: AWS, Linux Shell Scripting
Database: MySQL, Firebase
Web & Software Development: Flutter, Django, PHP
Document Creation: Microsoft Office Suite, Markdown, $\text{\LaTeX}$
Languages: Bengali (Native), English (IELTS Academic 8.5), French (Preparing for A1)

PROJECTS

TakaTracker – Expense Tracking Mobile Application 2024
Developed using Flutter and Firebase with features like currency conversion and categorized expense tracking

Cloud HemoSphere – Cloud-Based Blood Banking Web App 2023
Hosted on AWS using EC2 and VPC to ensure data security for blood bank management

IUBComp – Competition Portal Prototype 2023
Winning project at IUB Summer Tech Fest; image uploads, voting, and user submissions for university clubs

HONORS AND AWARDS

Mitacs Globalink Research Award January 2025
Competitive award for conducting research internship in Canada; Grant Amount: \$6000 CAD

3-Minute Thesis Competition Winner March 2024
1st Place at the CSE Annual Research Day, Independent University, Bangladesh

Microprocessor Project Showcase Champion December 2023
1st Place at IUB Autumn Tech Fest, recognized for technical innovation and presentation

Web Application Project Showcase Winner July 2023
1st Place at IUB Summer Tech Fest for developing an innovative competition platform prototype

Dean's List 2023
Awarded for academic excellence at Independent University, Bangladesh

Dean's Merit List 2022
Recognized for being among the top-performing students in the School of Computer Science

Merit Scholarship 2019–2022
Awarded based on university entrance exam; received full or partial tuition coverage across multiple semesters

WORKSHOPS

Interdisciplinary Computational Biology Workshop 2025 January 2025
3-Day Workshop at Brac University Dhaka, Bangladesh

2nd CCDS Workshop on Deep Learning Code Management December 2024
2-Day Workshop by CCDS, Independent University, Bangladesh Dhaka, Bangladesh